# Sachit Mahajan

sachitmahajan.sm@gmail.com | https://sachitm.github.io | linkedin.com/in/sachitmahajan

## SKILLS

**Programming**: C++, Python, C, Embedded C, MATLAB

**Software and Framework**: Pytorch, ROS/ROS2, PCL, Eigen, Gazebo, Isaac Sim, RTOS, OpenCV, Ceres, GTSAM, Git, Docker

## EXPERIENCE

**Amazon.com Services LLC** – SDE (I → II → III) , Fauna Robotics *Sunnyvale, CA | Jun 2021 – Present*

- Driving next-generation spatial perception and embodied AI R&D for general-purpose humanoid robots, and building scene representations for **3D spatial, temporal, and semantic reasoning** in long-term autonomy.
- Built a **3D Gaussian Splatting** based digital twin pipeline that renders robot trajectories in real homes to provide ground truth poses, dense stereo depth, and object semantic labels; scaling data for training and evaluation.
- Fine-tuned and evaluated learned visual-navigation policies (**residual RL, visual navigation transformer**) for a humanoid robot in Isaac Sim and Lab, with **1.5-2x** the success of an A*/Nav2 baseline around obstacles.
- Developed and shipped a **SLAM** module for Amazon's consumer robots, deployed to **thousands** of customers.
- Designed and deployed a **robust pose graph optimization (RPGO)** system to correct odometry drift, increasing stability and map accuracy by over **2.5x** by rejecting outliers. Optimized compute performance under hardware constraints.
- Built SLAM benchmarking pipelines, a long-term autonomy simulation framework that simulates realistic perturbations in a house, metrics framework and deterministic replay tools for root-cause analysis, improving stability metrics by **2x**.
- Built an AI-driven evaluation framework for non deterministic autonomy behaviors with fewer runs (LLM-as-a-Judge, statistical testing) and measuring sim2real gap; a golden-set selection method cut evaluation time by **70%**.

**Amazon.com Services LLC** – SDE Intern, Consumer Robotics (CoRo) *Remote | May 2020 – Aug 2020*

- Developed planar feature extraction and tracking (walls, floors, ceilings) to improve visual-inertial odometry in low-texture indoor environments, and integrated planar constraints into the SLAM pipeline, reducing relative pose error (RPE).
- Designed a geometric-prior loop closure validation module that reduced false positives in place recognition.

**GreyOrange Pte Ltd.** – R&D Embedded Engineer, Robotics *Gurgaon, India | Oct 2017 – Jul 2019*

- Developed embedded firmware for warehouse Autonomous Mobile Robots on an RTOS, covering navigation, system diagnostics, obstacle detection, and predictive emergency response modules for safe robot behavior.

## EDUCATION

**Carnegie Mellon University (CMU)** – School of Computer Science – Robotics Institute *Pittsburgh, PA | May 2021*

MS in Robotic Systems Development (MRSD)

**Vellore Institute of Technology (VIT)** *Vellore, India | July 2013 - May 2017*

B.Tech, Electronics & Communication Engineering

## PUBLICATIONS

[1] **POp-GS: Next Best View in 3D-Gaussian Splatting with P-Optimality** (***CVPR 2025***), Joey Wilson, Marcelino Almeida, **Sachit Mahajan**, Martin Labrie, Maani Ghaffari, Omid Alizadeh, Min Sun, Cheng-Hao Kuo, Arnie Sen.

[2] **Modeling Uncertainty in 3D Gaussian Splatting through Continuous Semantic Splatting** (***ICRA 2025***), Joey Wilson, Marcelino Almeida, Min Sun, **Sachit Mahajan**, Maani Ghaffari, Parker Ewen, Omid Alizadeh, Cheng-Hao Kuo, Arnie Sen.

[3] **Design of a Medical Assistant Robot** (***RTEICT 2017***), **Sachit Mahajan**, C.M. Vidhyapathi.

**Reviewer** for IEEE RA-L, IROS, ICRA, Humanoids, AAAI, CoRL

## PROJECTS

[1] **Navigation, Payload Handling and Docking, and Safety Behaviors for an Autonomous Vehicle ( )**: Developed localization, navigation, and precision-docking behaviors for an Autonomous Vehicle operating with modular payloads in geo-fenced environments.

[2] **SALSA - Semantic Assisted Lifelong Structure Aware SLAM ( ):** Learning-augmented lifelong SLAM using semantic landmarks to classify dynamic/repeatable objects. Incorporated high-confidence structural anchors to robustly localize with fewer observations.

[3] **Handling Mirrors in Indoor LiDAR Mapping**: Built a LiDAR-camera pipeline to detect mirrors and correct point-cloud maps, using a novel coarse-to-fine region-growing algorithm for mirror boundary estimation and artifact correction.